

Roadblock Examples for Each Root-Cause Category

For each of the thirteen validated root-cause categories in the taxonomy (grouped into eight families by the deciding technique and whether it *helps*, suffix **P**, or *hurts*, suffix **A**, roadblock resolution), we list one concrete roadblock from the labeled dataset: the *gate* it guards, the *mechanism* by which the technique helps or hurts, and the measured *evidence* (the tool and its named metric).

I2S-P — input-to-state substitution helps

`string_literal_substitution`

$n = 199$

Example. `1cms/2246` (`cmsChannelsOfColorSpace`) — a direct equality check against the 4-byte ICC colour-space signature 'CMYK' (`cmsSigCmykData`).

Mechanism. `cmplog` observes 'CMYK' in the comparison trace and substitutes it into the input in one step; random mutation essentially never produces the exact four bytes.

Evidence. `operand_enrichment`: `target_enrich` $\approx +17.9$ — 'CMYK' fills $\approx 24\%$ of the `cmplog` corpus at the gate offset vs. 0% of naive's.

`numeric_tag_substitution`

$n = 39$

Example. `bloaty` (`ElfMachineToCapstone`) — a flat `switch` over the 2-byte ELF header `e_machine`; case `EM_AARCH64` requires the field `== 183`.

Mechanism. I2S captures the numeric constant 183 straight from the comparison trace and substitutes it in one step; context-sensitive coverage offers no help because the control flow is flat with no path depth to reward.

Evidence. `operand_enrichment`: `target_enrich` > 1.0 (the magic value is enriched in the `cmplog` corpus). Same tool and rule as `string_literal_substitution`, split only because the operand is numeric rather than a string.

I2S-A — input-to-state substitution hurts

`target_depletion`

$n = 25$

Example. `libpng/3977` — a sub-field at a fixed input offset must equal the target byte 0x00.

Mechanism. naive lands 0x00 by ordinary havoc; `cmplog` harvests the abundant ASCII float-exponent 'e' (0x65) from other numeric comparisons and keeps splicing it into the same position, pinning the corpus onto that decoy and starving the target value.

Evidence. `operand_enrichment`: `target_enrich` $= -0.82$, `decoy_enrich` $= +0.68$ — the decoy fills 63% of the `cmplog` corpus (vs. 39% for naive) while the target drops to 8.9% (vs. 15.7%).

`decoy_substitution_overfit`

$n = 14$

Example. `harfbuzz/5323` (`hb_script_get_horizontal_direction`) — a `switch` on the decoded script; case `HB_SCRIPT_THAANA` fires only for Thaana text (U+0780..07BF).

Mechanism. naive produces short low-structure blobs that occasionally decode to exotic scripts; I2S keeps splicing the harvested OpenType table tags GPOS/GSUB into offsets 12–15, holding the font structurally valid and ASCII/Latin — so `cmplog` reaches the `switch` every trial but never produces the Thaana content the gate needs.

Evidence. `operand_enrichment`: `target_enrich` $= -2.79$, `decoy_enrich` $= +11.3$ — the GPOS/GSUB decoy fills 51% of the `cmplog` corpus vs. 0.05% of naive's.

`structural_byte_corruption`

$n = 10$

Example. harfbuzz/5831 (initial_reordering_consonant_syllable) — the gate needs a Kannada syllable spanning ≥ 3 glyphs (a long, repeated multi-byte glyph run).

Mechanism. naive builds the long run by havoc; I2S overwrites bytes inside the run with harvested constants, shortening it below the 3-glyph threshold, so the cmlog corpus is depleted of the assembled structure. The displacement is diffuse — no single dominant decoy.

Evidence. operand_enrichment: target_enrich = -13.6 (saturated: essentially erased at the operand offset) — the structural byte is present in 36% of naive seeds but only 20% of cmlog’s, with only diffuse displacement (3% vs. 0.8%).

VP-P — the value-profile gradient helps

gradient_value_distance_closure

$n = 33$

Example. lcms/2288 (cmsChannelsOfColorSpace) — a switch on the 4-byte ICC colour-space signature; case cmsSigMCH2Data needs all four exact bytes ‘MCH2’.

Mechanism. value_profile’s per-byte Hamming-distance feedback scores near-matches, so it keeps a partial hit and climbs the remaining bytes to the target; naive has no signal distinguishing “one byte right” from “none right” and stalls.

Evidence. value_distance_reached: the non-VP arm stalls at min_distance = 0.75 (one of four bytes right) while value_profile reaches 0.0 — it closes the whole gap.

gradient_assembly_depth

$n = 20$

Example. harfbuzz/5302 — a downstream gate that only larger / more token-dense valid fonts reach.

Mechanism. extra valid structure opens more comparison gradients downstream, so value_profile’s feedback rewards seeds that carry it and the whole corpus drifts toward bigger valid inputs; naive has no reason to grow its seeds and never accumulates the structure the gate needs.

Evidence. seed_size_and_tags: size lift = 4.69 — VP seeds average 258 B vs. 55 B, carrying a much larger valid font body at comparable tags.

VPC-P — input-to-state and value-profile are jointly necessary

joint_value_distance_closure

$n = 18$

Example. sqlite3/16001 — the gate resolves only when the input contains the exact 6-byte keyword ‘PRAGMA’ as a well-formed PRAGMA <name>; statement.

Mechanism. value_profile alone climbs toward the keyword (‘SELE...’) but never spells it; I2S alone keeps malformed tokens on which even a spliced ‘PRAGMA’ earns no coverage and is dropped. Only with both enabled does the gradient retain a near-keyword input onto which I2S then splices the exact literal.

Evidence. value_distance_reached: either technique alone stalls at min_distance ≈ 0.83 (the closest single-arm seed is five of six bytes off); only the joint arm lands it.

joint_assembly_depth

$n = 9$

Example. harfbuzz/6814 — a gate behind interlocking fields that needs a populated ‘CPAL’ colour table.

Mechanism. I2S places the exact table tag, but a single substitution pass does not grow the input far enough, so cmlog stalls at a partial structure; with both techniques enabled, vpc admits and progressively retains the near-complete inputs cmlog discards, growing the corpus to a structure deep enough to reach the gate.

Evidence. `seed_size_and_tags`: `value_profile` places no table tags (`vp_tags = 0`), `cmplog_tags = 1.4`, `vpc_tags = 2.4`; the vpc median seed length is $2.44\times$ that of the cmplog corpus.

CTX-P — context-sensitive coverage helps

`iteration_path_depth`

$n = 14$

Example. `libxml2/6467 (xmlParseCharRef)` — a decimal char-reference digit loop whose gate `count++ > 20` is reached only by inputs with more than 20 digits.

Mechanism. `naive`'s saturated edge map marks the loop edge covered on the first iteration and discards deeper seeds; `naive_ctx` records each edge with its call context, so every extra iteration counts as new coverage and the mid-progress seed is retained, letting the corpus extend past the threshold.

Evidence. `depth_reach`: `hit_ratio = 7.26` — `naive_ctx` runs the gate line $89.7\times$ per seed vs. `naive`'s $12.4\times$, and takes the blocked side $7.0\times$ more.

CTX-A — context-sensitive coverage hurts

`corpus_inflation`

$n = 4$

Example. `libpng/4072 (png_read_IDAT_data)` — a branch that needs residual compressed bytes left after zlib returns `Z_STREAM_END`, behind a deep inflate `do/while` loop.

Mechanism. `naive_ctx` folds a context hash into each of the loop's ≈ 2750 revisits, so every revisit looks like new coverage; it hoards a corpus far larger than `naive`'s and spends its mutation budget re-saving the same control flow instead of manufacturing the extra-data condition.

Evidence. `corpus_size_ratio`: `corpus_ratio = 3.73` — `naive_ctx` saves 16,761 corpus entries vs. `naive`'s 4,500.

GRIM-P — grammar-aware recombination helps

`structural_assembly`

$n = 11$

Example. `harfbuzz/6181 (hb_ot_shaper_categorize)` — a script dispatch whose `case HB_SCRIPT_GRANTHA` fires only for a structurally valid multi-table font.

Mechanism. `grimoire` splices learned fragments into a coherent multi-table font (`glyph`, `GSUB`, `GPOS`, `cmap`, `head`, `name`); `cmplog` can splice the one 4-byte table tag it logged but cannot recombine the several coherent constructs the gate needs.

Evidence. `seed_size_and_tags`: `structural_token_ratio  $\approx 6.1$` — `grimoire` carries 5.5 distinct structural tokens per seed vs. `cmplog`'s 0.9.

NGRAM-P — path-history coverage helps

`sequential_depth_reach`

$n = 2$

Example. `libxml2/6325 (UTF16LEToUTF8)` — a UTF-16 decode loop whose high-surrogate gate (`c & 0xFC00`) `== 0xD800` is reached only after decoding enough code units in order.

Mechanism. each pass through the decode region produces a fresh 4-edge path tuple, so `naive_ngram4` keeps the seed that gets one step further in, while plain `naive` sees the same block, calls it covered, and drops it; the ngram corpus drifts toward inputs that drive the decoder deep enough to reach the surrogate-assembly lines.

Evidence. `depth_reach`: `hit_ratio = 1.64` — the winner drives the blocker line $156.7\times$ per seed vs. $95.5\times$, into the surrogate-assembly lines the naive corpus never reaches.